

Progression of liver pathology in patients undergoing the Fontan procedure: Chronic passive congestion, cardiac cirrhosis, hepatic adenoma, and hepatocellular carcinoma.

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Staged palliative surgical procedures have been an effective treatment of complex congenital heart defects. The Fontan procedure has been of particular benefit to infants with functional single-ventricle complexes but with the consequence of a sustained increase of right-sided venous pressure. We reviewed the clinical and pathologic features of 9 autopsied patients having undergone the Fontan procedure, with special attention given to their liver pathology. The 9 patients died from a few hours to 18 years after the Fontan operation. Chronic passive congestion was seen in 7 patients, and 4 patients surviving 4 to 18 years also had cardiac cirrhosis. Hepatic adenoma in the setting of cardiac cirrhosis was found in a patient surviving for 9 years. One patient surviving for 18 years had hepatocellular carcinoma superimposed on cardiac cirrhosis. Rupture of the hepatoma in this case led to fatal hemorrhage. The study shows that chronically increased hepatic venous pressure from the Fontan procedure might lead to chronic passive congestion, cardiac cirrhosis, hepatic adenoma, and hepatocellular carcinoma.

The Australia and New Zealand Fontan Registry: description and initial results from the first population-based Fontan registry.

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The Fontan procedure is the final in a series of staged palliations for single-ventricle congenital heart disease, which encompasses rare and heterogeneous cardiac lesions. It represents an unusual and novel physiological state characterised by absence of a subpulmonary ventricle. The population is growing steadily, prompting creation of this registry to study their epidemiology, demographic trends, treatment and outcomes. This multicentre, binational, prospective and retrospective, web-based registry involving all congenital cardiac centres in the region has identified nearly all Fontan patients in Australia and New Zealand. Patients identified retrospectively were approached for recruitment. New recipients are automatically enrolled prospectively unless they choose to opt-out. Follow-up data are collected yearly. Baseline data were obtained in 1072 patients as at 1 January 2011. Ninety-nine patients died; 64 were lost to follow up. Forty-four per cent of patients lost were between 20 and 30 years of age. The size of the Fontan population is increasing steadily. Among 973 living patients, 541 (56%) gave consent for prospective collection of follow up. Between 1 January 2011 and 1 January 2013, an additional 47 subjects were enrolled prospectively. The current proportion of patients operated with hypoplastic left heart syndrome is currently 29% and is growing rapidly. The population surviving after the Fontan procedure has been growing in recent decades, especially since survival with hypoplastic left heart syndrome has improved. The Australia and New Zealand Fontan Registry provides population-based data, and only large databases like this will give opportunities for understanding the population and performing prospective trials.

Cardiac Imaging in Patients After Fontan Palliation: Which Test and When?

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The Fontan operation represents the final stage of a series of palliative surgical procedures for children

born with complex congenital heart disease, where a "usual" biventricular physiology cannot be restored. The palliation results in the direct connection of the systemic venous returns to the pulmonary arterial circulation without an interposed ventricle. In this unique physiology, systemic venous hypertension and intrathoracic pressures changes due to respiratory mechanics play the main role for propelling blood through the pulmonary vasculature. Although the Fontan operation has dramatically improved survival in patients with a single ventricle congenital heart disease, significant morbidity is still a concern. Patients with Fontan physiology are in fact suffering from a multitude of complications mainly due to the increased systemic venous pressure. Consequently, these patients need close clinical and imaging monitoring, where cardiac exams play a key role. In this article, we review the main cardiac imaging modalities available, summarizing their main strengths and limitations in this peculiar setting. The main purpose is to provide a practical approach for all clinicians involved in the care of these patients, even for those less experienced in cardiac imaging.

The Fontan Procedure for Single-Ventricle Physiology.

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The Fontan procedure is the final procedure in staged palliation for patients with functional single-ventricle physiology. The goal of the procedure is to separate systemic and pulmonary blood flow by directing systemic venous return through the Fontan connection to the pulmonary arteries and the lungs without ventricular contribution. Following the procedure, pulmonary blood flow is completely passive and dependent on pressure gradients, resulting in complex postoperative cardiopulmonary interactions. Understanding the physiology is essential to effectively manage these patients. Critical care nurses caring for patients after a Fontan procedure must understand preoperative data, risk factors, and unique postoperative physiology so they can anticipate specific postoperative problems, recognize trends in clinical status, and develop an appropriate plan of care. This paper reviews the first 2 stages of single-ventricle palliation, relevant modifications to the Fontan procedure, important preoperative cardiac catheterization data, common postoperative problems, and outcomes after the Fontan procedure.

Left pulmonary artery reconstruction using cryopreserved pulmonary homograft.

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In patients with univentricular heart, the Fontan procedure is the final palliation. This is usually staged. A systemic-to-pulmonary artery shunt is performed in the presence of episodes of cyanotic spells in the neonatal period or in some patients in infancy; a bidirectional superior cardiopulmonary anastomosis is preferred early in life. This is followed by the final Fontan palliation on an elective basis later. For an effective bidirectional superior cavopulmonary anastomosis and Fontan palliation, good-sized confluent pulmonary arteries (PAs) are mandatory in addition to favorable hemodynamic data. Patients with discontinuous PAs that are small in size present a surgical challenge at initial palliation as the one described in this report.

Rare problems associated with the Fontan circulation.

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The Fontan operation, originally described for the surgical management of tricuspid atresia, is now the final surgery in the strategy of staged palliation for a number of different forms of congenital cardiac disease with a functionally univentricular heart. Despite the improved technical outcomes of the Fontan operation, staged palliation does not recreate a normal physiology. Without a pumping chamber delivering blood to the lungs, the cardiovascular system is less efficient; cardiac output is generally diminished, and the systemic venous pressure is increased. As a result, patients with "Fontan physiology" may face a number of rare but potentially life-threatening complications including hepatic dysfunction, abnormalities of coagulation, protein-losing enteropathy, and plastic bronchitis. Despite the staged palliation resulting in remarkable survival, the possible complications for this group of patients are complex, involve multiple organ systems, and can be life threatening. Identifying the mechanisms associated with each of the rare complications, and developing strategies to treat them, requires the work of many people at many institutions. Continued collaboration between sub-specialists and between institutions will be required to optimise the care for this group of survivors with functionally univentricular hearts.

Occurrence of a primary liver cancer with an unusual histologic appearance as a late Fontan complication.

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The Fontan procedure is an open heart procedure performed in pediatric patients with a particular congenital cardiac anomaly known as a univentricular heart. The procedure is used to reroute the systemic venous blood from the inferior vena cava directly to the pulmonary artery. It improves patients' prognoses, but various late-phase extracardiac complications that manifest when patients reach adolescence have been recognized. These complications, pulmonary arteriovenous fistula and protein losing gastroenteropathy, for example, present significant challenges in the management of adults with Fontan circulation. Liver fibrosis is another possible late-phase complication and one of the most serious. Development of a neoplasm, usually a hepatocellular carcinoma, is sometimes reported. We encountered a young patient in whom Fontan circulation led to the development of a histologically unusual liver cancer that resembled the poorly differentiated hepatocellular carcinoma or the combined hepatocellular-cholangiocarcinoma with stem-cell features described in the latest WHO classification.

Fontan procedure: imaging of normal post-surgical anatomy and the spectrum of cardiac and extracardiac complications.

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Univentricular congenital heart diseases include a range of entities that result in a functionally single ventricular chamber. Although the only curative therapy is cardiac transplantation, there are several palliative surgical techniques that prevent ventricular volume overload, diverting part or all the systemic venous circulation into the pulmonary arteries. The modern Fontan procedure, which consists of anastomosing both the superior (SVC) and inferior vena cava (IVC) to the right pulmonary artery (RPA), is nowadays the last step before transplantation. The importance of imaging in these entities lies not only in the understanding of the new circuit established after surgical correction, but also in the early

detection of the wide spectrum of cardiac and extracardiac complications that can occur due to the new physiological condition. Due to the increased survival of these patients, long-term complications are becoming more common. The main cardiac complications are atrial enlargement, ventricular dysfunction, and stenosis or thrombosis of the conduit. Pulmonary artery stenosis, pulmonary arteriovenous fistulae (PAVF), systemic-pulmonary veno venous shunts (VVS), hepatic congestion, cardiac cirrhosis, and protein-losing enteropathy are potential extracardiac complications.

Efficiency differences in computational simulations of the total cavo-pulmonary circulation with and without compliant vessel walls.

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The Fontan operation is a palliative surgical procedure performed on children born with congenital defects of the heart that have yielded only a single functioning ventricle. The total cavo-pulmonary connection (TCPC) is the most popular variant of the Fontan procedure. The objective of the study was to quantify and compare the efficiency of numerical models of the TCPC with rigid versus elastic vessel wall models. The pressure drop and power loss through both type TCPC models was measured. Significant differences in efficiencies exist between rigid versus elastic numerical models. We have shown incorporating elasticity into numerical models of the total cavo-pulmonary connection is important when determining circuit efficiencies.

Fontan circulation and implications for future reproduction.

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This article summarizes and critiques four recent publications looking at preconception counseling, pregnancy outcomes, and cardiac complications in women with history of Fontan circulation. The Fontan procedure is a palliative strategy for single-ventricle type congenital heart disease and involves passive flow of venous return into the pulmonary circulation, bypassing the ventricles. Pregnancy in these patients is not without risk and preconception counseling and contraception practices vary widely. High rates of miscarriage, prematurity, and small-for-gestational-age babies are reported. Cardiac complications include mainly arrhythmias. Whether long-term prognosis in these patients is affected by pregnancy is not yet known.
